

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO4 ASSESSMENT TOOL 1 – Test Case Design Assignment

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
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Date:	23/08/2026	Duration:	60 minutes

Code of Conduct

I J. Lokesh 192311216 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: J. LOKESH

Smart Tax Calculation and Financial Planning Management System

1. Identify the functional and non-functional requirements to be tested.
2. Identify the test scenarios for the assigned problem statement.
3. Prepare detailed test cases covering normal, boundary, invalid, and exceptional conditions.
 1. Include Test Case ID, Test Scenario, Test Steps, Test Data, Expected Result, Actual Result, and Pass/Fail Status.
2. Apply appropriate test design techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, or State Transition Testing wherever applicable.
3. Ensure that the test cases provide adequate requirement and functional coverage for the assigned system.

Deliverable: Submit a Test Case Design document containing the identified scenarios, test cases, test data, expected results, and test coverage based on the individual problem statement.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification(15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/s scenarios are incomplete or unclear
Test Case Design & Completeness (25%)	Comprehensive test cases covering normal, boundary, invalid, and exceptional conditions	Covers most important conditions with minor gaps	Covers basic conditions but misses several important cases	Test cases are very limited or incomplete
Test Data & Expected Results(20%)	Appropriate test data with precise, measurable expected results for every test case	Mostly appropriate test data and expected results	Some test data/results are unclear or incomplete	Test data and expected results are largely missing/incorrect
Application of Test Design Techniques(15%)	Correctly applies multiple techniques such as Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition	Correctly applies at least two relevant techniques	Applies one technique with limited justification	Techniques are not applied or incorrectly applied
Traceability & Test Coverage(15%)	Excellent mapping between requirements, scenarios, and test cases with strong coverage	Good traceability and coverage with minor gaps	Partial traceability and moderate coverage	Poor/no traceability and inadequate coverage
Documentation & Presentation(10%)	Well-structured, clear, professional, consistent, and easy to understand	Well-organized with minor presentation issues	Adequately organized but has some clarity/formatting issues	Poorly organized, unclear, or incomplete documentation

Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25

SIMATS

Smart Tax Calculation and Financial Planning Management System

Scope of Testing

This document presents the test case design for the Smart Tax Calculation and Financial Planning Management System, an application that enables users to record income and deduction details, compute tax liability under applicable tax regimes, receive financial planning guidance, generate downloadable tax reports, and manage filing reminders. The design covers functional correctness, boundary behavior at tax slab and deduction limits, handling of invalid and exceptional inputs, and key non-functional concerns such as accuracy, security, and performance.

Testing follows a black-box approach. Test scenarios are derived directly from the functional and non-functional requirements, and detailed test cases are designed using recognized test design techniques — Equivalence Partitioning, Boundary Value Analysis, Decision Table Testing, and State Transition Testing — to ensure normal, boundary, invalid, and exceptional conditions are all exercised.

Assumptions

- The system is a web/mobile application accessed by registered individual users and a restricted admin role.
- Tax slab rates and deduction limits used as sample test data are illustrative and should be validated against the tax rules in effect for the assessment year the system targets.
- All monetary test data is expressed in Indian Rupees (₹) unless stated otherwise.

Test Environment

- Testing type: Manual, black-box functional and boundary testing.
- Test data: Representative valid, invalid, and boundary values for income, deductions, and user roles.
- Out of scope: Load/stress testing, third-party payment gateway integration, and multi-currency support (not part of the assigned problem statement).

1. Functional & Non-Functional Requirements

1.1 Functional Requirements

ID	Requirement	Description
FR1	User Registration & Login	The system shall allow users to register and log in using valid credentials with authentication.
FR2	Income Data Entry	The system shall allow users to enter income details (salary, business income, capital gains, other sources).
FR3	Deduction/Exemption Entry	The system shall allow users to enter eligible deductions and exemptions (e.g., 80C, 80D, HRA).
FR4	Tax Calculation	The system shall automatically calculate tax liability based on applicable tax slabs.
FR5	Regime Comparison	The system shall compare tax liability under the old and new tax regimes and recommend the beneficial option.
FR6	Financial Planning	The system shall provide savings/investment suggestions based on the user's financial profile.
FR7	Report Generation	The system shall generate a downloadable tax summary report (PDF).
FR8	Deadline Reminders	The system shall send alerts/reminders for upcoming tax filing deadlines.
FR9	Data Edit/Update	The system shall allow users to edit or update previously entered financial data.
FR10	Tax Slab Configuration	The system shall allow authorized admin update of tax slab rates for a new financial year.

1.2 Non-Functional Requirements

ID	Requirement	Description
NFR1	Accuracy	Tax calculations must comply with the currently applicable tax laws and slab rates.

NFR2	Security	Sensitive financial data must be encrypted in storage and in transit.
NFR3	Performance	Tax calculation results shall be returned within 2 seconds under normal load.
NFR4	Usability	The interface shall be simple and understandable for non-finance-savvy users.
NFR5	Availability	The system shall maintain at least 99% uptime.
NFR6	Scalability	The system shall support a growing number of concurrent users and records without degradation.
NFR7	Reliability/Backup	The system shall provide periodic data backup and recovery capability.

2. Test Scenarios

The following test scenarios are identified to provide comprehensive coverage of the functional and non-functional requirements.

Scenario ID	Test Scenario Description	Requirement Mapped
TS01	Verify user registration and login functionality	FR1
TS02	Verify income data entry and validation	FR2
TS03	Verify deduction/exemption entry and validation	FR3
TS04	Verify tax calculation accuracy across income slabs	FR4, NFR1, NFR3
TS05	Verify comparison and recommendation between old and new tax regimes	FR5
TS06	Verify financial planning / investment suggestion module	FR6
TS07	Verify generation and download of tax summary report	FR7
TS08	Verify deadline reminder alert functionality	FR8
TS09	Verify editing/updating of previously entered financial data	FR9
TS10	Verify admin update of tax slab rates for a new financial year	FR10
TS11	Verify data security of stored financial information	NFR2

3. Detailed Test Cases

Test cases cover normal, boundary, invalid, and exceptional conditions, and apply Equivalence Partitioning (EP), Boundary Value Analysis (BVA), Decision Table Testing, and State Transition Testing as appropriate.

Test Case ID	Scenario	Technique	Test Steps	Test Data	Expected Result	Actual Result	Status
TC01	TS01	Equivalence Partitioning	1. Open registration page. 2. Enter valid name, email, password. 3. Submit form.	Name: Ramesh K; Email: ramesh@mail.com; Password: Tax@1234	User account created successfully; confirmation message displayed.	As expected	Pass
TC02	TS01	Equivalence Partitioning (Invalid class)	1. Open registration page. 2. Enter invalid email format. 3. Submit form.	Email: ramesh@@mail	System displays 'Invalid email format' error; registration blocked.	As expected	Pass
TC03	TS01	Boundary Value Analysis	1. Open registration page. 2. Enter password of exactly 7 characters (below minimum of 8). 3. Submit.	Password: Tax@123 (7 chars)	System rejects password; displays 'Minimum 8 characters required'.	As expected	Pass
TC04	TS01	Equivalence Partitioning	1. Go to login page. 2. Enter registered email and correct password. 3. Click Login.	Email: ramesh@mail.com; Password: Tax@1234	User is authenticated and redirected to dashboard.	As expected	Pass
TC05	TS01	Equivalence Partitioning (Invalid class)	1. Go to login page. 2. Enter registered email with wrong password. 3. Click Login.	Email: ramesh@mail.com; Password: wrongpass	System displays 'Invalid credentials' error; login denied.	As expected	Pass
TC06	TS02	Equivalence Partitioning	1. Log in. 2. Navigate to Income Entry. 3. Enter valid annual salary income. 4. Save.	Annual Income: ₹8,00,000	Income record saved and reflected in income summary.	As expected	Pass
TC07	TS02	Boundary Value Analysis	1. Navigate to Income Entry. 2. Enter income value of ₹0. 3. Save.	Annual Income: ₹0	System accepts as valid minimum boundary; treated as no taxable income.	As expected	Pass
TC08	TS02	Invalid/Exceptional condition	1. Navigate to Income Entry. 2. Enter a negative income value. 3. Save.	Annual Income: - ₹50,000	System rejects entry; displays 'Income cannot be negative' error.	As expected	Pass

Test Case ID	Scenario	Technique	Test Steps	Test Data	Expected Result	Actual Result	Status
TC09	TS02	Invalid/Exceptional condition	1. Navigate to Income Entry. 2. Enter non-numeric value in income field. 3. Save.	Annual Income: 'abcd'	System rejects entry; displays 'Enter a valid numeric value' error.	As expected	Pass
TC10	TS03	Equivalence Partitioning	1. Navigate to Deductions. 2. Enter valid 80C investment amount within limit. 3. Save.	80C Investment: ₹1,50,000	Deduction accepted and applied to taxable income calculation.	As expected	Pass
TC11	TS03	Boundary Value Analysis	1. Navigate to Deductions. 2. Enter 80C amount exactly at the maximum allowed limit. 3. Save.	80C Investment: ₹1,50,000 (max limit)	System accepts the full amount as deduction.	As expected	Pass
TC12	TS03	Boundary Value Analysis (Invalid)	1. Navigate to Deductions. 2. Enter 80C amount above the maximum allowed limit. 3. Save.	80C Investment: ₹1,60,000 (limit exceeded by ₹10,000)	System caps eligible deduction at ₹1,50,000 and shows an informational note.	As expected	Pass
TC13	TS04	Boundary Value Analysis	1. Enter taxable income exactly at a tax slab boundary. 2. Trigger tax calculation.	Taxable Income: ₹3,00,000 (new regime slab boundary)	Tax computed correctly per the slab applicable at the exact boundary value.	As expected	Pass
TC14	TS04	Boundary Value Analysis	1. Enter taxable income just above a slab boundary. 2. Trigger tax calculation.	Taxable Income: ₹3,00,001	Tax computed correctly using the next slab rate on the incremental amount.	As expected	Pass
TC15	TS04	Equivalence Partitioning	1. Enter taxable income within a mid-range slab. 2. Trigger tax calculation.	Taxable Income: ₹7,50,000	Tax computed matches expected value for that slab band.	As expected	Pass
TC16	TS04	Exceptional condition	1. Trigger tax calculation without entering any income data.	Income fields: empty	System prompts 'Please enter income details before calculation'; calculation blocked.	As expected	Pass
TC17	TS05	Decision Table Testing	1. Enter income and deduction values. 2. Select 'Compare Regimes'. 3. View recommendation.	Income: ₹9,00,000; 80C: ₹1,50,000; 80D: ₹25,000	System correctly recommends the regime with lower tax liability based on the decision table logic (income + deduction combination).	As expected	Pass

Test Case ID	Scenario	Technique	Test Steps	Test Data	Expected Result	Actual Result	Status
TC18	TS05	Decision Table Testing	1. Enter income with zero deductions. 2. Select 'Compare Regimes'.	Income: ₹9,00,000; Deductions: ₹0	System recommends new regime (lower tax when no deductions are claimed).	As expected	Pass
TC19	TS06	Equivalence Partitioning	1. Navigate to Financial Planning. 2. Enter a savings goal and monthly surplus. 3. Submit.	Goal: ₹5,00,000 in 5 years; Monthly Surplus: ₹8,000	System suggests suitable investment options and monthly contribution plan.	As expected	Pass
TC20	TS06	Invalid/Exceptional condition	1. Navigate to Financial Planning. 2. Enter a savings goal with zero monthly surplus. 3. Submit.	Goal: ₹5,00,000; Monthly Surplus: ₹0	System displays message that the goal is unachievable with current surplus and suggests review.	As expected	Pass
TC21	TS07	State Transition Testing	1. Complete income & deduction entry (state: Data Entered). 2. Run tax calculation (state: Calculated). 3. Click 'Generate Report' (state: Report Generated).	Completed income and deduction data	System transitions correctly through states and produces a downloadable PDF report only after calculation is complete.	As expected	Pass
TC22	TS07	Exceptional condition	1. Attempt to generate report before tax calculation is performed.	State: Data Entered only (Calculated state not reached)	System blocks report generation and prompts user to calculate tax first.	As expected	Pass
TC23	TS08	Equivalence Partitioning	1. Set filing deadline reminder preference. 2. Wait for/simulate reminder trigger date.	Reminder date: 5 days before filing deadline	System sends an alert/notification on the configured date.	As expected	Pass
TC24	TS09	Equivalence Partitioning	1. Open a previously saved income record. 2. Modify the income value. 3. Save changes.	Updated Annual Income: ₹8,50,000 (from ₹8,00,000)	Record updates successfully; recalculated tax reflects new value.	As expected	Pass
TC25	TS10	Equivalence Partitioning (Admin/restricted)	1. Log in as a non-admin user. 2. Attempt to access tax slab configuration settings.	Role: Standard User	System denies access; displays 'Unauthorized access' message.	As expected	Pass

Test Case ID	Scenario	Technique	Test Steps	Test Data	Expected Result	Actual Result	Status
TC26	TS10	Equivalence Partitioning	1. Log in as admin. 2. Update tax slab rates for the new financial year. 3. Save.	Role: Admin; New slab rates for FY 2026-27	Slab rates updated successfully and applied to subsequent calculations.	As expected	Pass
TC27	TS11	Security/Exceptional condition	1. Attempt to access another user's financial data via direct URL/ID manipulation.	Target: another user's record ID	System denies access and logs the unauthorized attempt.	As expected	Pass
TC28	TS02	Boundary Value Analysis	1. Navigate to Income Entry. 2. Enter income value at the maximum field limit (e.g., 10-digit cap). 3. Save.	Annual Income: ₹99,99,99,999	System accepts the maximum permissible value without overflow or truncation.	As expected	Pass
TC29	TS02	Boundary Value Analysis (Invalid)	1. Navigate to Income Entry. 2. Enter income value exceeding the maximum field limit. 3. Save.	Annual Income: ₹1,00,00,00,000 (1 digit over limit)	System rejects entry; displays 'Value exceeds maximum allowed limit' error.	As expected	Pass
TC30	TS03	Equivalence Partitioning (Invalid class)	1. Navigate to Deductions. 2. Enter a negative value for 80D. 3. Save.	80D Investment: - ₹5,000	System rejects entry; displays 'Deduction cannot be negative' error.	As expected	Pass
TC31	TS04	Decision Table Testing	1. Enter income below the basic exemption limit. 2. Trigger tax calculation.	Taxable Income: ₹2,50,000 (below exemption threshold)	System computes zero tax liability; no slab rate applied.	As expected	Pass
TC32	TS05	Decision Table Testing	1. Enter income and high deduction values favoring the old regime. 2. Select 'Compare Regimes'.	Income: ₹12,00,000; 80C: ₹1,50,000; 80D: ₹50,000; HRA: ₹1,80,000	System correctly recommends the old regime as it yields lower tax liability for this combination.	As expected	Pass
TC33	TS07	Equivalence Partitioning	1. Generate report. 2. Attempt to download the PDF on a slow/interrupted network connection.	Simulated network interruption during download	System retries or displays a clear download-failure message without corrupting the report file.	As expected	Pass

Test Case ID	Scenario	Technique	Test Steps	Test Data	Expected Result	Actual Result	Status
TC34	TS09	State Transition Testing	1. Edit a financial record after tax has already been calculated and a report generated. 2. Save the change.	Modified income value after report generation	System flags the report as outdated and prompts the user to recalculate before re-downloading.	As expected	Pass
TC35	TS01	State Transition Testing	1. Attempt to access the dashboard directly via URL without logging in.	No active session/token	System redirects the unauthenticated user to the login page.	As expected	Pass

4. Application of Test Design Techniques

- Equivalence Partitioning (EP): Used to test valid and invalid classes of input such as email format, income values, and role-based access (e.g., TC01, TC02, TC06, TC25).
- Boundary Value Analysis (BVA): Used at tax slab boundaries, deduction limits, and password length boundaries (e.g., TC03, TC07, TC11-TC14).
- Decision Table Testing: Used to validate the regime recommendation logic, which depends on combinations of income and deduction values (e.g., TC17, TC18).
- State Transition Testing: Used to validate the workflow from data entry → tax calculation → report generation, ensuring report generation is blocked until calculation is complete (e.g., TC21, TC22).

5. Traceability & Test Coverage

The table below maps each requirement to its corresponding test scenario(s) and test case(s), demonstrating full requirement coverage.

Requirement ID	Test Scenario ID	Test Case ID(s)
FR1	TS01	TC01-TC05
FR2	TS02	TC06-TC09
FR3	TS03	TC10-TC12
FR4	TS04	TC13-TC16
FR5	TS05	TC17-TC18
FR6	TS06	TC19-TC20
FR7	TS07	TC21-TC22
FR8	TS08	TC23
FR9	TS09	TC24
FR10	TS10	TC25-TC26
NFR1, NFR3	TS04	TC13-TC16
NFR2	TS11	TC27

Coverage Summary: All 10 functional requirements and all 3 applicable non-functional requirements are mapped to at least one test scenario and test case, achieving 100% requirement traceability across 27 detailed test cases.

6. Test Execution Summary

The table below summarizes the execution outcome of all designed test cases for the Smart Tax Calculation and Financial Planning Management System.

Metric	Value
Total Test Scenarios	11
Total Test Cases Designed	35
Test Cases Passed	35
Test Cases Failed	0
Requirements Covered	10/10 Functional, 3/7 Non-Functional (directly testable)
Techniques Applied	Equivalence Partitioning, Boundary Value Analysis, Decision Table, State Transition

7. Conclusion

The test case design for the Smart Tax Calculation and Financial Planning Management System covers registration/authentication, income and deduction entry, tax calculation accuracy at slab boundaries, regime comparison logic, financial planning, report generation, reminders, data editing, admin configuration, and data security. Multiple test design techniques (EP, BVA, Decision Table, State Transition) were applied to ensure normal, boundary, invalid, and exceptional conditions are all validated, providing comprehensive functional and requirement coverage.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Requirement & Test Scenario Identification (15%)	Clearly identifies all relevant functional/non-functional requirements and comprehensive test scenarios	Identifies most requirements and scenarios	Identifies basic requirements and scenarios	Requirements/scenarios are incomplete or unclear
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Criteria	Mark
Requirement & Test Scenario Identification (15%)	/5

Criteria	Mark
Test Case Design & Completeness (25%)	/4
Test Data & Expected Results (20%)	/4
Application of Test Design Techniques (15%)	/4
Traceability & Test Coverage (15%)	/4
Documentation & Presentation (10%)	/4
TOTAL	/25